

Cormac Delaney

AI Engineer | Agent Systems & Fintech

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PROFESSIONAL SUMMARY

AI engineer with 5+ years building and operating agent-based systems in production fintech environments. Proven track record shipping multi-agent orchestration platforms serving thousands of concurrent users and billions in transaction volume. Deep expertise in LLM integration, API orchestration, and cross-functional collaboration with product, compliance, and customer success teams. Currently focused on reducing operational friction through autonomous systems in payments workflows.

EXPERIENCE

Senior AI Systems Engineer | Pinnacle Financial Automation | San Francisco, CA | Jan 2024 – Present

- Architected and deployed a multi-agent transaction reconciliation system that processes 450M+ transactions monthly, reducing manual reconciliation overhead by 68% and cutting settlement delays from 2 days to 4 hours
- Implemented agentic tool-calling framework for dynamic API selection across 12 payment processors (Stripe, ACH networks, international rails); agents autonomously route transactions based on cost optimization and regulatory constraints
- Designed persistent memory layer for agent context, enabling agents to maintain decision history and regulatory audit trails across sessions; integrated with PostgreSQL and Redis for sub-100ms lookups at scale
- On-call 2 weeks per month; led incident response for agent hallucination event affecting 8K merchant accounts, implemented safeguards for financial amount validation, deployed fix within 90 minutes
- Presented agent architecture and risk controls to C-level stakeholders and institutional customers during quarterly business reviews; secured buy-in for expansion budget

AI Platform Engineer | Cascade Payments | Portland, OR | Aug 2022 – Dec 2023

- Shipped customer dispute resolution agent to production, handling 150K+ disputes monthly with 89% autonomous resolution rate; integrated with merchant APIs, customer portals, and Stripe's dispute management API
- Implemented function calling logic for agent to retrieve transaction history, contact customer communication logs, and apply merchant policies; built monitoring dashboards tracking agent confidence scores and human escalation rates
- Iterated on agent behavior post-launch based on merchant feedback; conducted 6 in-person customer interviews to understand dispute workflows, incorporated learnings into agent prompts and tool definitions
- Reduced human agent workload by 40 FTE; owned metrics tracking and weekly reporting to VP of Operations
- Built integration connectors to 8 third-party payment networks and compliance platforms; managed API versioning and backward compatibility as external systems evolved

ML Engineer | Waypoint Analytics | Denver, CO | Jun 2021 – Jul 2022

- Developed recommendation agent for fraud prevention that integrates real-time transaction data, historical patterns, and external risk feeds; deployed to 25K+ SMB merchant accounts
- Deployed using FastAPI and cloud infrastructure (AWS ECS, Lambda); managed containerization and CI/CD pipelines
- Collaborated with compliance team to ensure model decisions met regulatory requirements; documented reasoning chains for audit purposes

TECHNICAL SKILLS

Agent & LLM: Tool calling, prompt engineering, orchestration frameworks, memory management, function composition

Fintech: Payment APIs, settlement workflows, KYC/AML, dispute resolution, transaction reconciliation

Infrastructure: Python, FastAPI, AWS (ECS, Lambda, RDS), PostgreSQL, Redis, Docker, Terraform

Integrations: Stripe, ACH networks, Plaid, Sift Science, custom webhook systems

EDUCATION

B.S. Computer Science | University of Colorado | Boulder, CO | 2019